

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 01-02-2026**

Time: 02.00Pm to 05.00Pm **GTA-13** **Max. Marks: 180**

01-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P2)_GTA-13_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2025-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 8)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 9 – 16)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 25 – 32)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 40)	Questions with Multiple Correct Choice with partial mark	+4	-2	4	16
Sec – IV(Q.N : 41 – 48)	NUMERICAL VALUE	+4	0	8	32
Total				**16**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

1. If $\vec{a},\vec{b},\vec{c}$ are three non-coplanar unit vector, each inclined with other at angle $\frac{\pi}{6}$, then volume of tetrahedron whose edge are $\vec{a},\vec{b}$ and $\vec{c}$ is :

A) $\frac{\sqrt{3\sqrt{3}-5}}{12}$ **B)** $\frac{2\sqrt{3}+5}{12}$ **C)** $\frac{3+5\sqrt{2}}{12}$ **D)** None

2. Let $S_r=\frac{r^5+3r^4+2r^3+2}{r(r+1)(r+2)}$ *then* $\left[\sum_{r=1}^{10} S_r\right]=$

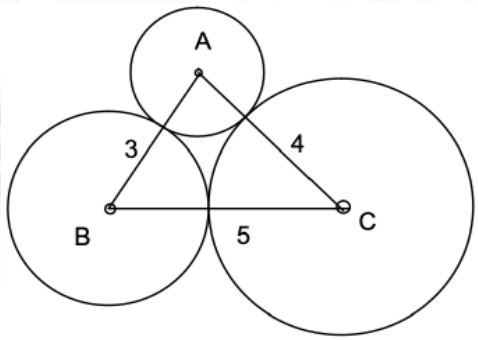
(Where [.] represent Greatest Integer Function)

A) 386 **B)** 385 **C)** 388 **D)** 390

3. Let $y=y(x)$, $y(1)=1$ and $y(e)=e^2$. Consider $J=\int\frac{x+y}{xy}dy, I=\int\frac{x+y}{x^2}dx$, $J-I=g(x)$ and $g(1)=1$, then the value of g(e) is

A) 3e + 1 **B)** e + 1 **C)** e^2-e+2 **D)** e^2+e-2

4. The vertices of a right triangle with sides (3, 4, 5) are the centres of three mutually externally tangent circles, as shown. What is the sum of the area of the three circles ?



A) 14π **B)** $\frac{25\pi}{2}$ **C)** 13π **D)** $\frac{27\pi}{2}$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

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Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Let $A=\begin{bmatrix}1 & 2\\ -a^2 & a+2\end{bmatrix}$ and B, C are 2×2 matrices such that $ABC=\begin{bmatrix}4 & 2\\ 1 & 3\end{bmatrix}$. Let $(BC)^T=[p_{ij}]_{2\times 2}$ and $p_{21}=-1$, then the value of a can be

A) 0 **B)** $-\frac{3}{2}$ **C)** $\frac{1}{2}$ **D)** $-\frac{1}{2}$

6. Consider, $f(x)=\tan^{-1}\left(\frac{\sqrt{1-4x^2}-2\sqrt{3}x}{\sqrt{3-12x^2}+2x}\right)$ which of the following alternative(s) is/are correct ?

A) If $x\in\left(\frac{-\sqrt{3}}{4},\frac{1}{2}\right]$ then range of f(x) is $\left[\frac{-\pi}{3},\frac{\pi}{2}\right)$

B) If $x\in\left[\frac{-1}{2},\frac{-\sqrt{3}}{4}\right)$ then range of f(x) is $\left(\frac{-\pi}{2},\frac{-\pi}{3}\right]$

C) The value of $f\left(\frac{\sqrt{3}+1}{4\sqrt{2}}\right)+f\left(\frac{1}{2\sqrt{2}}\right)$ is equal to $\frac{-\pi}{3}$

D) The value of $f\left(\frac{\sqrt{3}+1}{4\sqrt{2}}\right)+f\left(\frac{1}{2\sqrt{2}}\right)$ is equal to $\frac{-\pi}{12}$

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7. Consider a real valued function $H(x)=\int_{-1}^{-1-x^2}(z+2)(z+5)(z+10)dz$, then

A) H(x) has 7 critical points

B) $H''(x)=0$ has exactly 3 positive roots

C) $H'''(x)=0$ has exactly 2 positive roots

D) $H'(x)H''(x)=0$ has 13 distinct real roots

8. If S_n denotes the sum of n terms of the series $(1\le n\le 9)1+22+333+.....+\underbrace{999....9}_{9\,times}$ then

for $n\le 9$

A) $S_n-S_{n-1}=\frac{1}{9}\left(10^n-n^2+n\right)$ **B)** $S_n=\left(10^n-n^2+2n-2\right)$

C) $9\left(S_n-S_{n-1}\right)=n\left(10^n-1\right)$ **D)** $S_3=356$

SECTION-III
(NUMERICAL VALUE)

- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

9. Let f(x) is a polynomial such that $f(x)+f''(x)=x^3+7x$, then

$$\lim_{n\to\infty}\frac{\sum_{r=1}^{n}f(2r)}{1.2^2+2.3^2+....+n(n+1)^2}$$ is equal to

10. Let x and y be two real numbers such that $2\sin x\sin y+3\cos y+6\cos x\sin y=7$, then

$\tan^2 x+\tan^2 y$ is equal to

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11. Let z_1 *and* z_2 are two distinct roots of the equation $z^{101}=1$ then the probability that $|z_1+z_2|\geq\sqrt{2+\sqrt{3}}$ is

12. Let $f(x)=\begin{cases}|x|^{p-2}\sin\frac{1}{x}+x|\tan x|^{q-3} & x\neq 0\\ 0 & x=0\end{cases}$, is differentiable at x = 0, then minimum value of [p + q] equals (where [.] represents greatest integer function)

13. Let y = f(x) be a curve which satisfy the differential equation $e^x dy+(x-1)dx=0$ and $f(0)=0$. If the area enclosed by the curve, x-axis, y-axis, and a line x = c, where (c, f(c)) is the point of inflection on the curve is $\left(1-\frac{p}{e^q}\right)(p,q\in N)$, then the value of $(p+q)$ equals

14. If the range of values of 'a' for which $f(x)=\log_a(4ax-x^2)$ us strictly increasing $\forall x\in\left[\frac{3}{2},2\right]$ is $(p,q]\cup(r,\infty)$ then the value of (2p + 4q + r) equals

15. Given the equation of the ellipse $\frac{(x-3)^2}{16}+\frac{(y-4)^2}{49}=1$, a parabola is such that its vertex is the lowest point of the ellipse and it passes through the ends of the minor axis of the ellipse. The equation of the parabola is in the form $16y=A(x-H)^2-K$. Determine the value of $\frac{A}{7}+\frac{H}{3}+\frac{K}{16}$ is equal to

16. The lines $L_1,L_2,L_3,.....L_{20}$ are distinct. All the lines L_4,L_8,L_{12},L_{16} *and* L_{20} are parallel. All the lines $L_1,L_5,L_9,L_{13},L_{17}$ pass through a given point A. The maximum number of points of intersection of these 20 lines is.

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PHYSICS **Max Marks: 60**

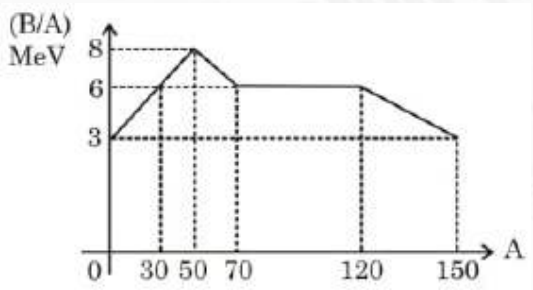
SECTION–I
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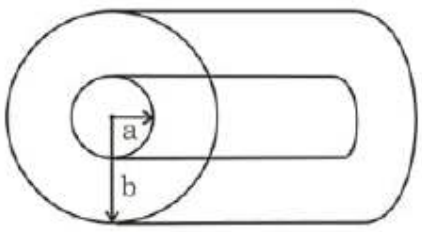
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17. Assume that the nuclear binding energy per nucleon (B/A) versus mass number (A) is as shown in the figure. Consider a nucleus of A = 110. Fission of this nucleus results into 2 fragments. Which of the following could possibly be the mass number of the resulting nuclei ?



A) 80 and 30 **B)** 60 and 50 **C)** 100 and 10 **D)** 90 and 20

18. We have a solid cylindrical rod made of aluminium having a hallow portion filled with iron. The left flat end and right flat end are maintained at constant but different temperature. The curved surface is insulated from surroundings and also from each other. The rate of heat flow through both rods is equal. The thermal conductivity of aluminium is 240 w/mK and that of iron is 80 w/mK. Then b/a =



A) 3 **B)** 1.5 **C)** $\frac{2}{\sqrt{3}}$ **D)** $\frac{4}{3}$

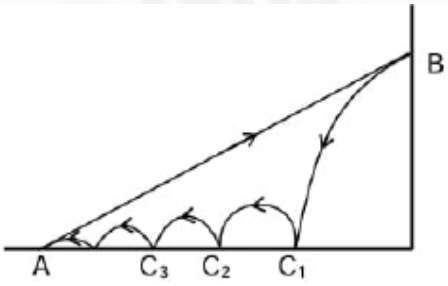
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19. An empty freight car on a level railroad track has a mass M. A chute above the freight car opens and grain falls down in to the car at a rate of 'r', measured in kilograms per second. The grain falls a vertical distance h before hitting the bed of the freight car without bouncing up. How much normal force is exerted on the freight car from the rails at a time t after the grain begins to hit the bed of the car ? Assume the grain starts from rest.

A) $Mg + rtg$ **B)** $Mg + r\sqrt{gh}$

C) $Mg + r\sqrt{gh} + rtg$ **D)** $Mg + r\sqrt{2gh} + rtg$

20. A ball was projected from point A and it hits a vertical wall at point B horizontally. From point B the ball goes to points $C_1, C_2, \ldots\ldots$ and finally gets back to point A. The coefficient of restitution is e at any point. Find the possible value of e, if the ball does not bounce up at all when it gets back to point A.



A) $e = \sqrt{2} - 1$ **B)** $e = \sqrt{3} - 1$ **C)** $e = \dfrac{\sqrt{3}-1}{2}$ **D)** $e = \dfrac{1}{\sqrt{2}}$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions

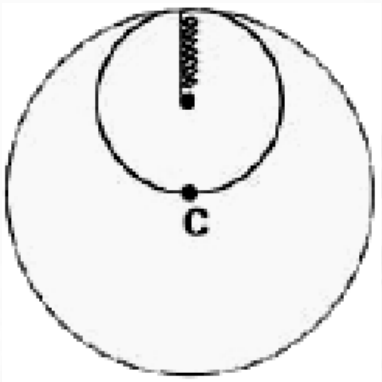
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21. A non-conducting sphere having a spherical cavity centred at a distance 'a' from centre of sphere. Charge inside the sphere is distributed uniformly with volume density ρ. A particle of mass m having charge –q is suspended vertically from light non-conducting spring of force constant k. Choose the correct statement(s). (there is no gravity and initially when –q is suspended, the spring was in natural length)



A) Maximum elongation of the spring is $\left(\frac{2\rho aq}{3\varepsilon_0 k}\right)$

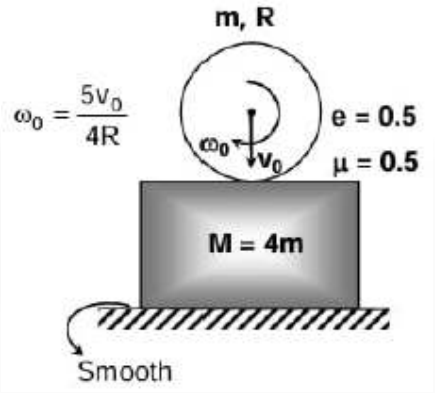
B) Maximum elongation of the spring is $\left(\frac{\rho aq}{3\varepsilon_0 k}\right)$

C) Time period of oscillation is $2\pi\sqrt{\frac{m}{k}}$

D) Time period of oscillation is $2\pi\sqrt{\frac{\rho m}{\varepsilon_0 k}}$

22. A solid sphere of mass m and radius R having initial angular velocity $\omega_0 = \frac{5v_0}{4R}$ and initial velocity v_0 collides with a plank of mass M = 4m with a coefficient of restitution e = 0.5 as shown in the figure. The plank is placed on a smooth horizontal surface. The coefficient of friction between the sphere and the plank is $\mu = 0.5$. Then choose the correct option(s)

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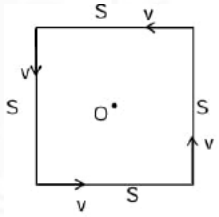
A) The angular velocity of sphere just after collision is $\frac{5v_0}{12R}$

B) The angular velocity of sphere just after collision is $\frac{5v_0}{6R}$

C) The velocity of plank just after collision is $\frac{v_0}{12}$

D) The impulse due to friction on the sphere during collision is $\frac{mv_0}{3}$

23. Four particles are located at the vertices of a square of side 'a'. They all start moving simultaneously with constant speed v with the first particle always heading for the second, the second for the third, the third for the fourth and the fourth for the first. If 'τ' is the time in which particles converge, then choose the correct option(s)



A) The distance covered by each particle till they converge is a

B) The acceleration of each particle at $t = \frac{\tau}{5} is \frac{5v^2}{4a}$

C) The acceleration of each particle at $t = \frac{\tau}{5} is \frac{5v^2}{2a}$

D) The radius of curvature of the trajectory of each particle at $t = \frac{\tau}{5} is \frac{2a}{3}$

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24. A string of mass m is fixed at both of its ends. The fundamental mode of string is excited and it has an angular frequency ω and maximum displacement amplitude A. Then :

A) The maximum kinetic energy of the string is $E_k = \frac{1}{4}mA^2\omega^2$

B) The maximum kinetic energy of the string is $E_k = \frac{1}{2}mA^2\omega^2$

C) The mean kinetic energy of the string averaged over tone period time is

$< E_k >= \frac{1}{4}mA^2\omega^2$

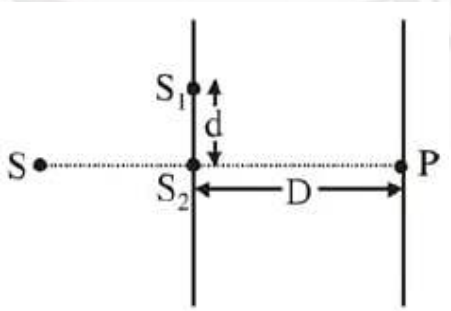
D) The mean kinetic energy of the string averaged over tone periodic time is

$< E_k >= \frac{1}{8}mA^2\omega^2$

SECTION-III
(NUMERICAL VALUE)

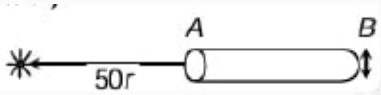
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25. In a YDSE experiment two slits S_1 *and* S_2 have separation of d = 2 mm. The distance of the screen is D = 8/5m. Source S starts moving from a very large distance towards S_2 perpendicular to S_1S_2 as shown in figure. The wavelength of monochromatic light is 500 nm. The number of maximas including at infinity observed on the screen at point P as the source moves Untill S_2 on the line S_2 P is 3995+n. Find the value of n.



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26. A cylindrical rod of length $l = 64$ cm and cross-sectional radius $r = \left(2/\sqrt{\pi}\right)$ cm is placed at a distance 50 r from a infrared point source S of power 1.25 kW as shown in the figure. The lateral surface of the rod is perfectly insulated from the surroundings. The cross-section A absorbs 80% of the incident energy, has temperature T_A in steady state. The surface B is radiating energy into space and the wavelength emitted by it with maximum energy density is 100,000 A. Determine the temperature of end A (in K)
(Wein's constant= 0.003 mK)



27. From a thin convex lens of focal length F = 20 cm and aperture diameter of 10 mm a small piece was cut as shown in the figure. The piece is then placed in contact with the remaining part of the lens on the original principal axis of the lens. If the optical centre of the original lens is taken to be origin of co-ordinates and a point object is placed at (-30, 0), then the image coordinate becomes (x, y), in centimeter find the value of x + y
(Consider image the formation by both lenses only)

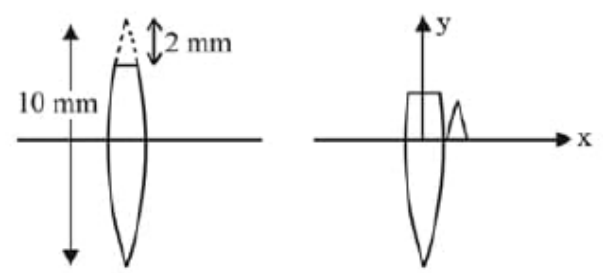
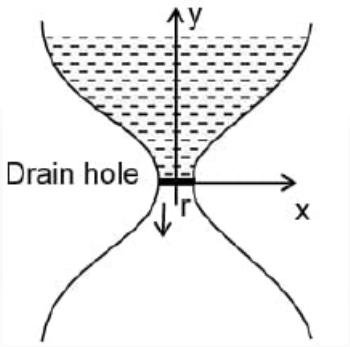


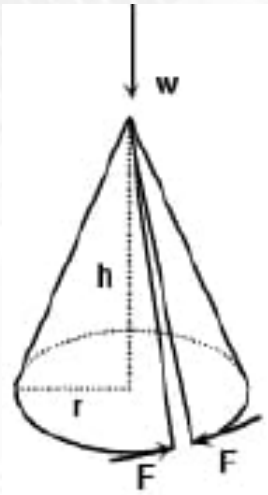
Figure-1 Figure-2

28. The shape of an ancient water clock jar is such that the water level descends at a constant rate at all times. The water level falls by λ m/s and the shape of the jar is given by $y \alpha x^n$. Find the value of $\left(\frac{3n}{5}\right)$, if the radius r of the drain hole can be assumed to be very small

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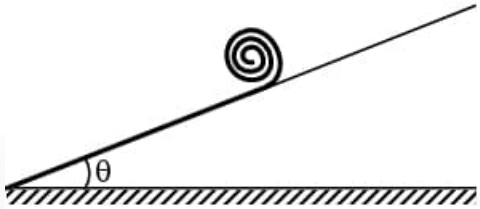


29. A cone with height h = 1 m and a base circle of radius r = 1 m is formed from a sector – shaped sheet of paper. The sheet is of such a size and shape that its two straight edges almost touch on the sloping surface of the cone. In this state the cone is stress-free. The cone is placed on a horizontal. Slippery table-top, and loaded at its apex with a vertical force of magnitude $w = 2\pi$, without collapsing. The splaying of the cone is opposed by a pair of forces of magnitude F acting tangentially at the join in the base circle (see figure). Ignoring any frictional or bending effects in the paper, find the value of F.

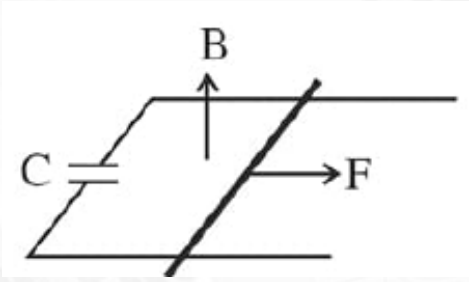


30. A carpet runner is placed onto a long straight inclined plane parallel to the steepest line in the plane, from the bottom of the slope till its top. The carpet is thin and flexible, and it cannot slide down because of the friction, but it can easily be lifted, it does not stick to the plane. The top end of the carpet is rolled, a bit, and this hoop of carpet is released. The hoop rolls down the slope with greater and greater speed, while its diameter increases, and finally it reaches the bottom of the slope during a time of t_1. If a solid cylinder is released at the top of the slope, it reaches the bottom in a time of t_2. The ratio of $\frac{t_1}{t_2} = \sqrt{k}$. Find k.

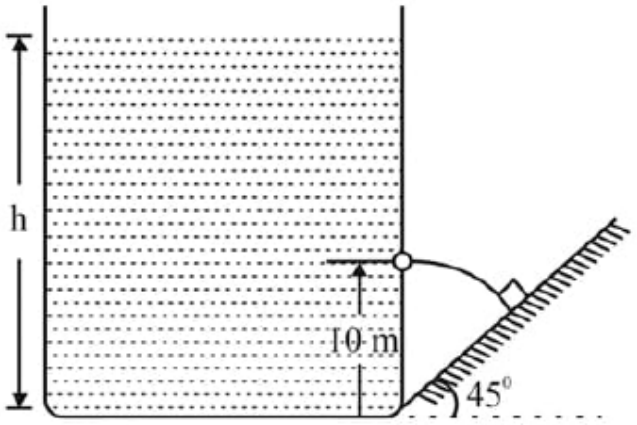
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31. A rod of resistance R, length l and mass m is placed on two smooth parallel rails which are resistanceless. The rails are connected by capacitor of capacitance C. A uniform magnetic field B is present in vertical direction. The rod is pulled at it's centre by a constant force F. If $m = B^2 l^2 C,\ RC = 2\text{sec}, F = 7.29N, B = 0.1T\ and\ l = 2m,$ what is current (in A) in rod at t = 2 sec (e = 2.7)



32. A rectangular tank of height h filled with water is placed near the bottom of a plane mirror inclined at an angle 45° with the horizontal. At height 10 m from bottom a hole is made such that steam coming out from hole strikes the plane mirror normally. Then at the time of hitting, speed of image of water droplet hitting the plane mirror, formed in plane mirror (in m/s) is $(Take\, g = 10\, m/s^2)$



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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

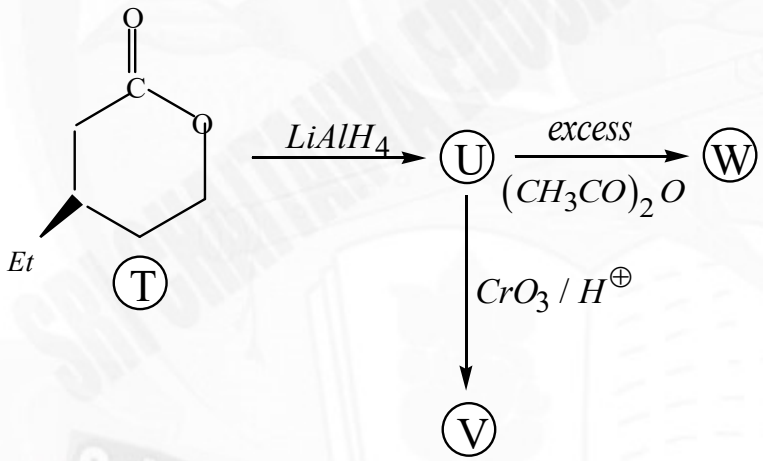
33. Which complex absorbs visible light with maximum wave length ?

A) $\left[Co(SCN)_6\right]^{3-}$ **B)** $\left[Co(CN)_6\right]^{3-}$ **C)** $\left[Ir(SCN)_6\right]^{3-}$ **D)** $\left[Ir(CN)_6\right]^{3-}$

34. Stability of $\left[Cu(CN)_4\right]^{3-}$ and $\left[Cd(CN)_4\right]^{2-}$ can be best compared using ____

A) KCN(excess) **B)** H_2S **C)** $dil.H_2SO_4$ **D)** $NaOH\,(excess)$

35. With reference the scheme given, about T. U, V & W how many are correct:



i) 'T' is soluble in hot aq NaOH

ii) 'U' is optically active

iii) molecular formula of W is $C_{10}H_{18}O_4$

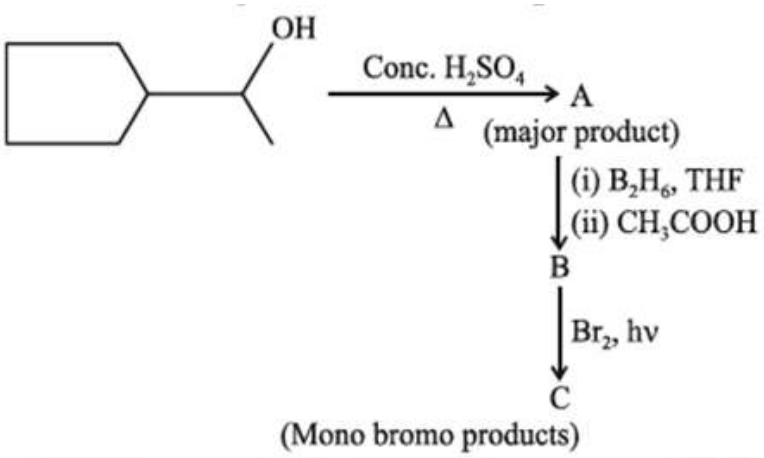
iv) V gives effervescence with aq $NaHCO_3$

from given statements(s) (i to iv)

A) 3 **B)** 1 **C)** 2 **D)** 4

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36. Consider the given reaction sequence:



i) Number of fractions obtained from fractional distillation of "C" is 8.

ii) "A" on reaction with Br_2 / CCl_4 can give three stereoisomers.

iii) Total number of mono-bromo products is 14.

iv) "B" is optically active.

From statement(s) (i to iv) regarding the above reaction sequence how many is/are correct

A) 1 **B)** 2 **C)** 3 **D)** 4

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.

For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

37. Which statements(s) is (are) correct ?

A) Average potential energy of two rotating polar bonds is more sensitive to intermolecular distance compared to stationary dipoles.

B) Oxygen is soluble in water through induced dipolar attractions.

C) Xenon gas is adsorbed on coconut charcoal through induced activated dipolar interactions.

D) In solid xenon the inter atomic forces are dependent to the 6th power of inter atomic distances.

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38. A "nonmetal-hydrogen" covalent bond (s) other than "O-H" is (are) developed during

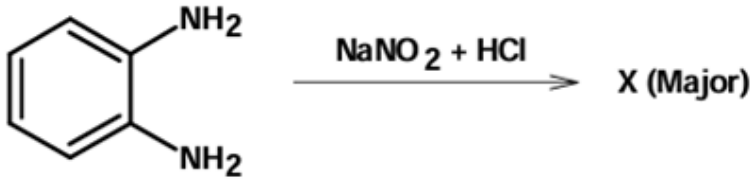
A) Action of NaOH(aq) on white phosphorous

B) Hydrolysis of PCl_3

C) Acid action on sodium sulphite

D) Hydrolysis of NCl_3

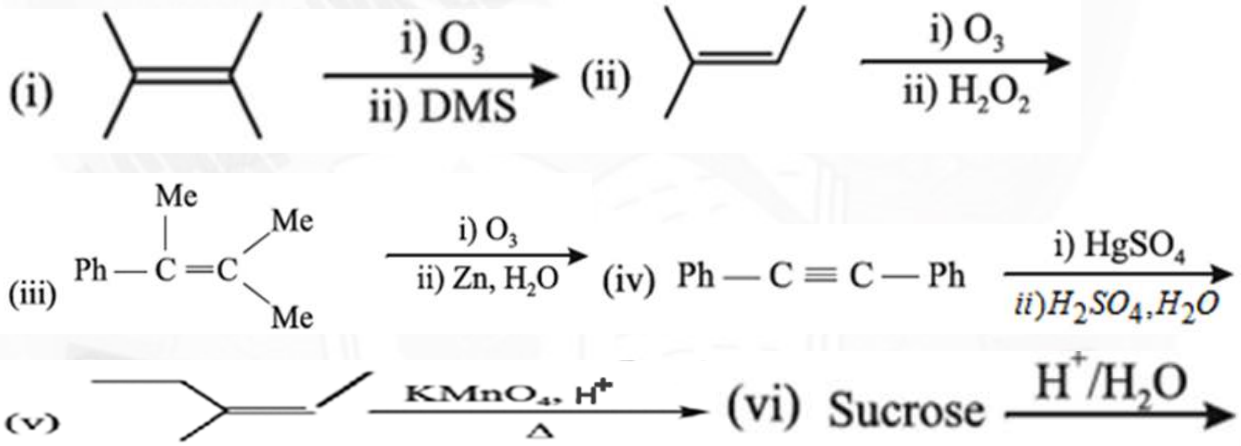
39. From following reaction



A) Degree of unsaturation of 'X' is 5 **B)** Degree of unsaturation of 'X' is 6

C) 'X' contains two rings **D)** 'X' contains one ring

40. From following reactions, select correct options(s)



A) Both the products of (ii) gives +ve response to Haloform reaction

B) Product of (i) is more reactive towards nucleophilic addition with methyl alcohol than product of (iv)

C) Products of (iii) and (v) gives positive response to Tollen's test

D) Only from product of (vi) gives positive response to Tollen's test

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SECTION-III
(NUMERICAL VALUE)

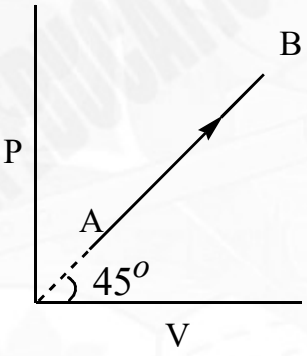
- This section contains **EIGHT (08)** questions stems.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +4*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

41. Maximum of how many grams of $MgCl_2$ can be dissolved in a 200 ml solution containing $0.2M\ NH_3$ and 0.1 M NH_4Cl without causing precipitation of $Mg(OH)_2$?

K_{sp} of $Mg(OH)_2 = 1.2\times10^{-12}$, K_b of NH_3 is 10^{-6}.

(log 2 = 0.3; At.wts : Mg = 24, Cl = 35.5)

42. A monoatomic ideal gas undergoes a hypothetical reversible path shown in the graph



What percentage of the supplied heat is converted into work?

43. The chloride salt (MCl) and Bromide salt (MBr) of a metal 'M' are mixed and washed with pure water and the residue is removed. The clear solution obtained is used in the following cell. $(E^o_{cell} = 1.0 Volts)$

$$M \,/\, \begin{matrix} MBr(sat) \\ MCl(sat) \end{matrix} \,//\, \begin{matrix} MBr(sat) \\ MCl(sat) \end{matrix} \,/\, \underset{(5.76atm)}{Cl_2}\ (g) : Pt$$

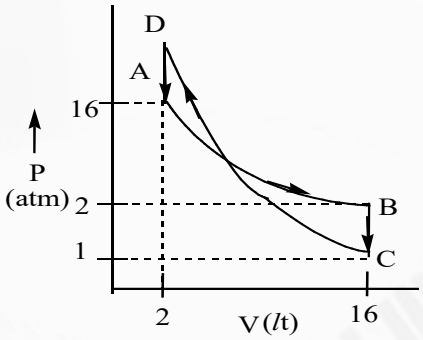
What is E_{Cell} of the given cell (in volts) ?

K_{sp} of MCl = 2.4×10^{-11}; K_{sp} of MBr = 4.0×10^{-11}, $\frac{2.303RT}{F} = 0.06$; $\log 2 = 0.3$

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44. An ideal gas undergoes the following reversible cyclic thermodynamic process $\left(C_V = \frac{3}{2}R\right)$ (graph not to scale)



$A \to B: isothermal$

$C \to D: Adiabatic$

How many of the following relations are correct?

a) $(\Delta S)_{BC} = (\Delta S)_{DA}$ b) $Q_{BC} = Q_{DA}$ c) $(\Delta S)_{AB} = -2(\Delta S)_{DA}$

d) $(\Delta T)_{BC} = (\Delta T)_{DA}$ e) $\frac{T_B}{T_C} = \frac{T_D}{T_A}$ f) $\left|nc_{vm} l\,\mathrm{n}\frac{T_D}{T_C}\right| = \left|nRl\,\mathrm{n}\frac{V_D}{V_C}\right|$

45. A 'one molal' Aluminum sulphate solution is prepared by consuming 171 grams of solute. The solution gives out 100 grams of ice at $'-x^oC'$. What is x ?
Crycoscopic constant of water is $2k\,kg\,mol^{-1}$. At wts : $Al = 27, S = 32, O = 16$
(Salt is 100% ionized).

46. Aqueous potassium formate is subjected to electrolysis (kolbe's) in order to generate that much Hydrogen gas which can hydrogenate 0.1 mole of "methylidene cyclobutene" into n-pentane. For how much time (minutes) electrolysis should be carried out in a 10 ampere circuit ? (1F = 96500C)

47. How many isomeric octahedral complexes can exist (theoretically) with the formula $Co(NH_3)_3(NO_2)_3$?

48. What could be the instantaneous deBroglie wave length $\left(A^0\right)$ of an electron at the moment where it has $\frac{1}{4\pi}A^o$ uncertainty in position having 15% error in measurement of velocity ?

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